

Table 1: Oracle threshold variation results for failure rate for Judge-based effectiveness fitness f_{dim} and efficiency fitness function $f_{efficient}$. Structured fitness f_{eff} threshold is kept 0.75 (balancing borderline fail vs. complete pass). Statistically significant differences using Wilcoxon signed-rank test (p-value below 0.05) comparing (LUNAR/S vs. SENSEI) are highlighted in bold. If results are not statistically significant, neither approach has superior performance. Suffix I represents open-source SUT and suffix II represents industrial SUT. Effect sizes are Large. Note: Less constraining higher thresholds (i.e., 0.9, 1.0) partially lead to large failure spaces making all approaches fail similarly.

(a) Carcontrol-I							
LLM	Approach	0.5	0.6	0.7	0.8	0.9	1.0
gpt-5-chat	LUNAR	0.274	0.312	0.313	0.313	0.313	0.313
	LUNAR_S	0.095	0.124	0.124	0.124	0.124	0.124
	SENSEI	0.005	0.042	0.062	0.062	0.096	0.096
gpt-4o	LUNAR	0.334	0.470	0.470	0.470	0.470	0.470
	LUNAR_S	0.112	0.231	0.233	0.233	0.233	0.233
	SENSEI	0.000	0.117	0.140	0.140	0.162	0.162
(b) Navi-I							
LLM	Approach	0.5	0.6	0.7	0.8	0.9	1.0
gpt-5-chat	LUNAR	0.574	0.591	0.597	0.597	0.597	0.597
	LUNAR_S	0.198	0.255	0.264	0.264	0.264	0.264
	SENSEI	0.013	0.051	0.339	0.339	0.390	0.390
gpt-4o	LUNAR	0.588	0.612	0.631	0.631	0.631	0.631
	LUNAR_S	0.283	0.330	0.341	0.341	0.341	0.341
	SENSEI	0.028	0.051	0.181	0.181	0.238	0.238
(c) Carcontrol-II							
LLM	Approach	0.5	0.6	0.7	0.8	0.9	1.0
gpt-5-chat	LUNAR	0.003	0.141	0.141	0.141	0.141	0.141
	LUNAR_S	0.003	0.171	0.171	0.171	0.171	0.171
	SENSEI	0.005	0.111	0.159	0.159	0.186	0.186
gpt-4o	LUNAR	0.005	0.115	0.115	0.115	0.115	0.115
	LUNAR_S	0.007	0.073	0.073	0.073	0.073	0.073
	SENSEI	0.000	0.000	0.000	0.000	0.004	0.004
(d) Navi-II							
LLM	Approach	0.5	0.6	0.7	0.8	0.9	1.0
gpt-5-chat	LUNAR	0.221	0.252	0.261	0.261	0.261	0.261
	LUNAR_S	0.185	0.236	0.237	0.237	0.237	0.237
	SENSEI	0.002	0.047	0.119	0.119	0.578	0.578
gpt-4o	LUNAR	0.225	0.305	0.305	0.305	0.305	0.305
	LUNAR_S	0.174	0.278	0.278	0.278	0.278	0.278
	SENSEI	0.002	0.106	0.231	0.231	0.524	0.524